

SniperFlow — Part 1 Research Report SniperFlow (XAU/USD only)

Part 1: Research Report Effective date: 05 Sep 2025

Introduction

This report evaluates three Android apps that overlap with the planned capabilities of SniperFlow: TradingView, FXStreet, and Myfxbook. The objective is to understand where each one excels, where gaps remain for an accuracy first, gold specific assistant and also which proven patterns we should adopt. Beyond describing features, we justify why particular ideas are innovative for our use case and explain likely Android implementation details (based on Android Studio practice), which informs our own design and build plan.

1) TradingView (Android)

Overview :

TradingView's Android app brings its powerful charting, watchlists, alerts and synced layouts to mobile. Users can draw, apply indicators and also receive push alerts tied to prices or strategies configured on web or mobile. It supports multiple assets, including commodities such as gold and integrates social "Ideas."

Strengths :

- Best in class charting on mobile: stable rendering across timeframes, smooth pan/zoom, and a large catalog of indicators. This matters for gold intraday analysis where precision of levels and responsiveness are crucial.
- Alerts and synchronization: push notifications mirror web setups, allowing traders to keep consistent triggers without duplicating work.
- Cross market coverage: manage FX, commodities, indices, and crypto in a single view; useful when tracking USD, yields, and gold together.

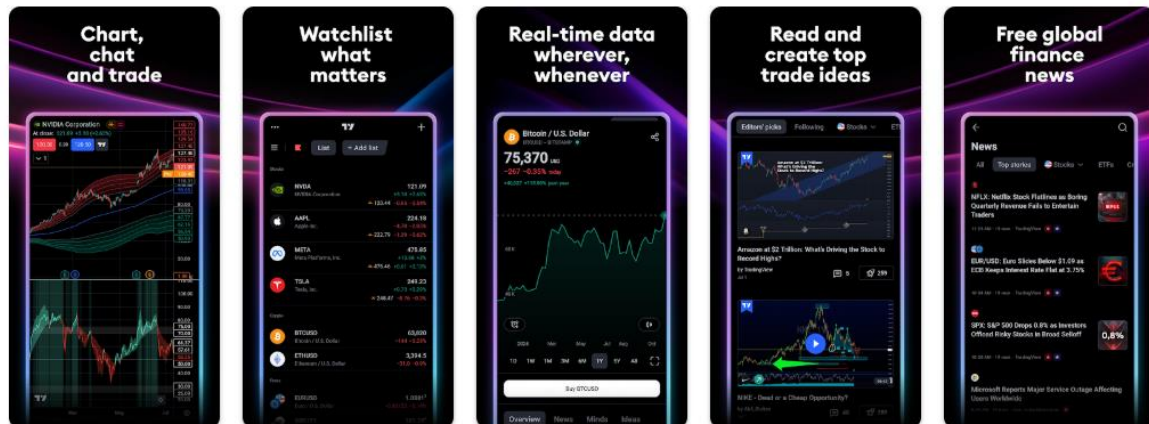
Weaknesses :

- Advanced strategy alerts and webhooks tend to require paid tiers; more importantly, TradingView does not explain "why" an alert should be trusted in the context of a trader's plan.
- Calendar/news exist but they are not gold centric and do not translate headlines/releases into a signed expected tilt for gold.
- Journaling is not a first class flow; offline capabilities centre on cached visuals rather than a deliberate capture now sync later mode.

How it was likely implemented (Android Studio view) A native Android shell hosts a high performance WebView for the chart (WebGL/JS), bridged to native toolbars, navigation, and notifications. State (watchlists, layouts) is cached locally with

encrypted preferences/SQLite; price/alert data is aggregated by TradingView's servers. FCM handles pushes. The app probably uses modular architecture with a repository pattern to abstract local/remote data.

Screenshots:



2) FXStreet (Android)

Overview :

FXStreet specialises in real time FX news and an economic calendar with impact levels. Each event includes previous values, consensus/forecast, and the actual number once released. The app highlights high impact events like US CPI, NFP, and FOMC, which are particularly relevant to gold's intraday moves.

Strengths

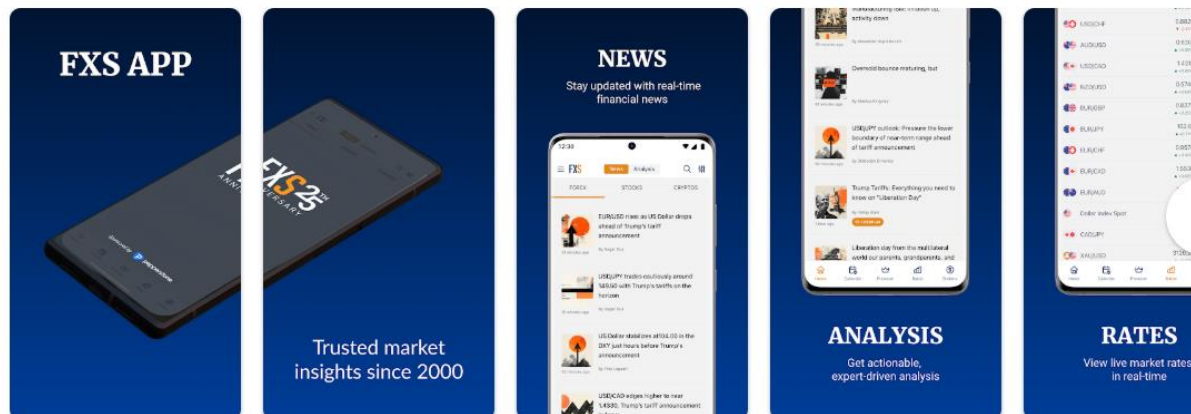
- Calendar UX: clear timing, impact tagging, and live updates at release time; filters and countdowns support pre release preparation and immediate post release interpretation.
- Timely headlines: expert analysis and curated news flow help users stay aligned with macro narratives.
- Integrated charts for quick checks, enough to visualise immediate reactions without leaving the app.

Weaknesses

- Personalisation is limited; the app is not aware of a user's specific plan. There is no built in notion of session gates, quiet hours, risk caps, or per user news mute windows.
- Headline/release effects on gold remain implicit; the app does not compute a signed tilt for XAU/USD.
- Advertising and information density can be distracting on a phone; the app is not designed around offline journaling.

How it was likely implemented (Android Studio view) Fully native list/detail screens backed by REST endpoints for calendar and news. The calendar likely uses paging and background refresh; push notifications are used for high impact items. A WebView or native chart view provides light charting. The data layer caches recent events/headlines with Room or a key value cache to keep the app responsive.

Screenshots:



3) Myfxbook (Android)

Overview :

Myfxbook is known for account analytics and verification, community outlook, and a portfolio of utilities including a live economic calendar, news, and pattern detection. The Android app aims to be a hub for portfolio tracking as well as market information.

Strengths

- Account analytics + verification increase trust and help traders evaluate performance over time.
 - Multiple utilities in one place: live calendar, markets, patterns, news, and alerts.
 - Community outlook gives sentiment context which some traders use as a contrarian input.
- Weaknesses
- The UI can be busy; ads and tool density reduce focus for a narrow, execution ready workflow.
 - Journal first thinking and explainability are not core; the app does not tell you why an alert matches a plan or what macro drivers matter now.
 - Offline support is basic; not tailored for deliberate offline capture and later sync.
- How it was likely implemented (Android Studio view) A group of native list/detail screens consuming Myfxbook APIs for accounts, calendar, and news. Notifications and ad SDKs are integrated. SQLite/Room is used for basic caching; business logic is centered on accounts rather than a plan aware trading flow.

Screenshots:



Comparison table :

App	Auth (Email/Password)	Google SSO	Push Alerts	Economic Calendar	News Feed	Chart Overlays	XAU/USD Focus
TradingView (Android)	✓	✓	✓	✓	✓	✓	—
FXStreet (Android)	—	—	✓	✓	✓	~	—
Myfxbook (Android)	✓	—	~	✓	✓	~	—
SniperFlow (MVP spec)	✓	✓	✓	✓	✓	✓	✓

Best features to adopt in SniperFlow :

From TradingView

- Smooth, accurate charts with reliable gesture handling; deep link from a push to the exact candle/time.
- Alert transport: we will accept TradingView webhooks as one of the inputs into our gating engine.

From FXStreet

- Impact tagging and clear countdowns; post release “Actual vs Forecast vs Previous” presentation with surprises highlighted.
- Economic release detail pages as an entry point to context.

From Myfxbook

- Compact “at a glance” market tiles and the idea of quick semantic tags—adapted to journals (e.g., “mss”, “sweep”, “news”).

Innovations SniperFlow contributes :

- Glass box Alert Detail: for any alert we show gate decisions (Session, News mute, Risk), a confidence meter, and a drivers list that itemises contributions (e.g., CPI surprise mapping to a small negative tilt lasting 60–90 minutes).
- Nowcast fusion: combine macro (real yields, USD index proxy, COT positioning) with intraday features (range regime, sweep/mss markers, prior day high/low context). Each signal contributes a signed delta with a time decay, producing a transparent “gold tilt” that the user can audit.
- XAU/USD first overlays: DO/PDH/PDL, Asia/London/NY killzones, and London Fix defaults that matter specifically for gold.
- Offline first journaling: create entries in <30 seconds during a live market and sync automatically when the network returns.
- Bilingual UI: English and isiZulu at launch, with Afrikaans strings added.

How the research shapes our build

- We will keep charts lightweight on mobile by using a WebView bridge for rendering but compute overlays/logic natively for clarity and accessibility.
- We will adopt FXStreet’s event fields and add our signed “gold tilt” effect to the Nowcast with a visible decay timer.
- We will intentionally avoid ads and dashboard clutter, focusing on the plan aware flow (alert → explain → chart → journal).

Conclusion

Generalist apps set high bars for charting (TradingView) and calendar/news cadence (FXStreet) but do not provide plan aware, explainable signals for gold or a journaling flow optimised for mobile and offline. SniperFlow will combine the best patterns, a precise mobile chart and a clean calendar with our unique contributions: glass box gating, nowcast fusion and also a rapid journal.

References:

- TradingView — App, T. (2025) *TradingView: Track all markets - apps on google play, Google*. Available at: <https://play.google.com/store/apps/details?id=com.tradingview.tradingviewapp&hl=en&pli=1> (Accessed: 05 September 2025).
- FXStreet — App, Fxs. (2025) *FXStreet – Forex & Crypto News - Apps on Google Play, Google*. Available at: <https://play.google.com/store/apps/details?id=com.fxstreet.forexnews&hl=en> (Accessed: 05 September 2025).

- Myfxbook — APP, M. (2025) *Forex calendar, Market & News - apps on Google Play, Google*. Available at: <https://play.google.com/store/apps/details?id=com.myfxbook.forex&hl=en> (Accessed: 05 September 2025).
- Data & APIs — ECONOMICS, T. (2025) *API - calendar: Trading economics, Api - Calendar | Trading Economics*. Available at: <https://tradingeconomics.com/calendar/api#:~:text=The%20Trading%20Economics%20API%20provides%20you%20with%20direct,like%20Microsoft%20Excel%20or%20a%20custom%20developed%20application>. (Accessed: 05 September 2025).
- ; API, F. (2025) *Federal Reserve Economic Data, St. Louis Fed Web Services: FRED® API*. Available at: <https://fred.stlouisfed.org/docs/api/fred/> (Accessed: 05 September 2025).
- ; COT, C.P.R. user guide (2025) *User's Guide, Go to publicreporting.cftc.gov*. Available at: <https://publicreporting.cftc.gov/stories/s/User-s-Guide/p2fg-u73y/> (Accessed: 05 September 2025).

Planning & Design Document

Introduction

SniperFlow is an Android app that helps gold traders act with discipline. It provides explainable bias (a Nowcast with visible drivers), “gated” alerts (only when sessions are active, news mute isn’t in effect, and risk limits allow), a TradingView compatible chart view and a fast offline first journal. This document details the features, UI, API design, storage, notifications, UML, data schema, and a 28 day plan to complete the MVP, explicitly covering all POE requirements.

Overview (name, icon, innovation)

- Name: SniperFlow (XAU/USD only MVP)

- Icon:



- Innovation: “glass box” Alert Detail and a Nowcast that merges macro (real yields, USD proxy, COT) with intraday features (range/vol, sweep/mss) to produce a signed tilt; bilingual (EN/ZU/AF); SAST aware sessions; offline first journal.

Detailed requirements (POE minimums + our additions)

Authentication & SSO

- Email/password via Firebase Auth (secure hashing managed by provider) plus optional custom token for our API. Google SSO for one tap sign in.
- Passwords are never stored locally; tokens in EncryptedSharedPreferences.

User settings

- Language (EN/ZU/AF) with instant apply.
- Sessions (Asia, London, New York) defaulted for SAST with edit & validation.

- Risk limits: trades/day, max daily loss %, optional soft cool off after SL.
- News mute presets: CPI, NFP, FOMC/Powell (e.g., +20/-40 minutes).
- Notifications: Plan Ready, Gate Pass/Blocked, Economic Release, News, Quiet hours.
- Data & About: Refresh macro, Clear cache, Privacy & Terms links.

REST API integration

Our backend normalizes external sources into simple endpoints. All responses include timestamps and a stale flag; auth via Bearer JWT.

- GET /price → { symbol, price, ts }
- GET /ohlc?tf=1m&from;=...&to;=... → array of { t,o,h,l,c }
- GET /features → intraday metrics (ATR20 multiple, range %, volume proxy)
- GET /levels/today → { do, pdh, pdl }
- GET /nowcast → { p_now, p_macro, p_final, contributors:[{name,delta}] }
- GET /calendar/upcoming → list of { id, timeSast, title, impact, forecast, previous, country }
- GET /calendar/release/{id} → adds actual, surprise, zScore, tiltDelta
- GET /news/headlines → { ts, source, title, url, topics[], sentiment, tiltDelta }
- POST /journal (auth) → persisted server side; queued offline in Room when no network. Upstreams (free tiers used responsibly)
- TradingEconomics (calendar/headlines), FRED (real yield series), CFTC (COT public reporting), and a free price/OHLC source.

We implement failover and serve stale behavior.

Offline mode with Sync

- Room caches OHLC (recent), alerts, levels, and journals.
- Journals saved offline get synced=false and enqueue via WorkManager.
- A global Offline banner appears; read only data is labeled "Cached".

Real time notifications

- FCM channels: plan (DEFAULT), alerts (HIGH), econ (HIGH), news (DEFAULT), system (MIN). Deep links: chart timestamp, alert detail, or event detail.

Multi language (South Africa)

- EN + isiZulu + Afrikaans string packs. Dates/times shown in SAST; currency US dollars.

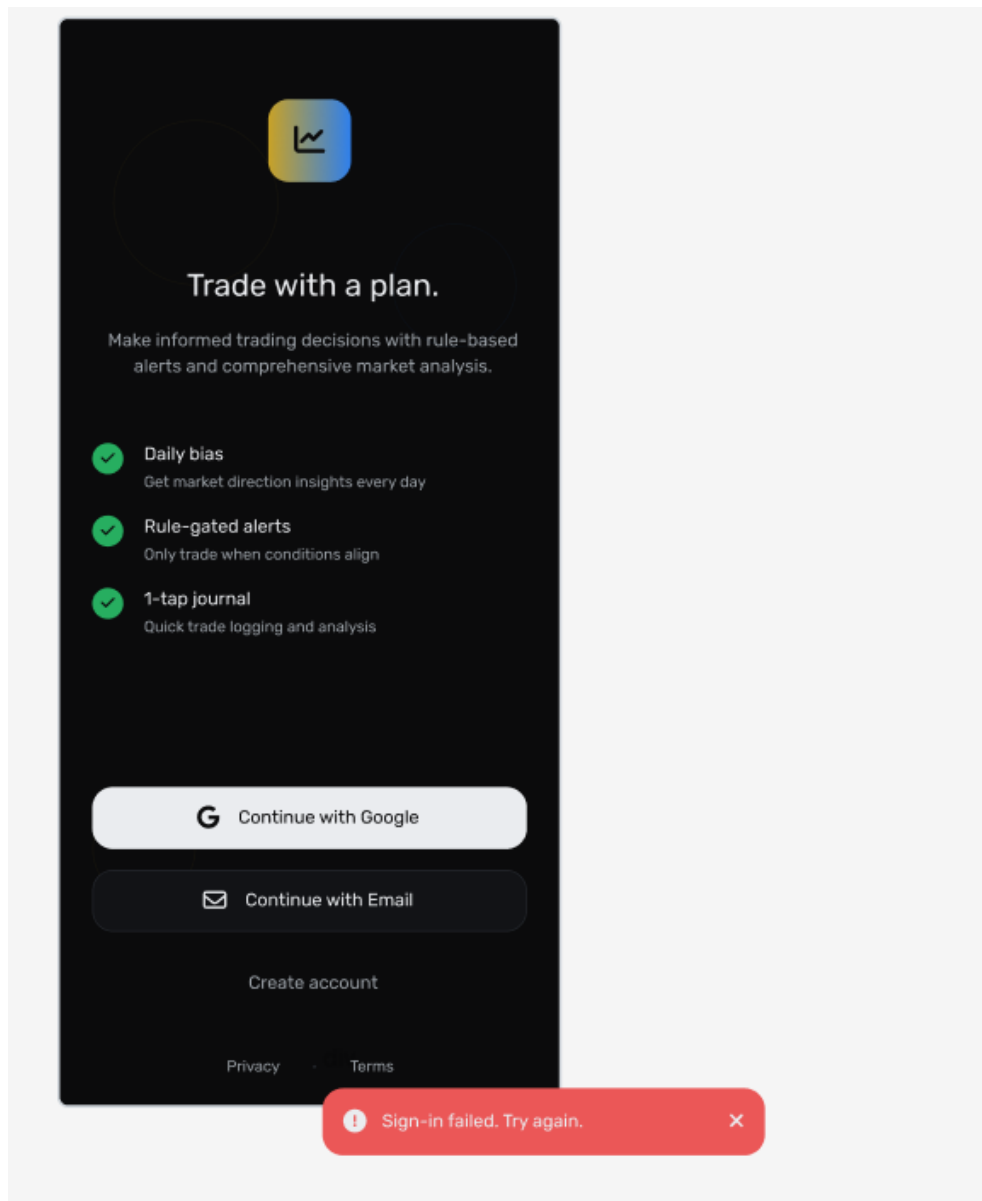
User interface design (screens & purpose)

Splash:



The splash screen features a dark background (matte black/charcoal) with a golden adaptive scope reticle at the center, symbolizing precision, focus, and the “sniper” identity of the app. The minimalist design emphasizes professionalism and accuracy, creating an immediate sense of trust and high-value intelligence. The gold accent stands out against the dark backdrop, reinforcing the app’s premium, trading-first positioning

Welcome:



The Welcome screen introduces users to SniperFlow with a modern, minimal dark theme and a centered gradient app icon.

The headline “Trade with a plan.” sets the tone, while a short description highlights the app’s purpose:

“Make informed trading decisions with rule-based alerts and comprehensive market analysis.”

Below the description, three core benefits are displayed with green checkmarks for quick impact:

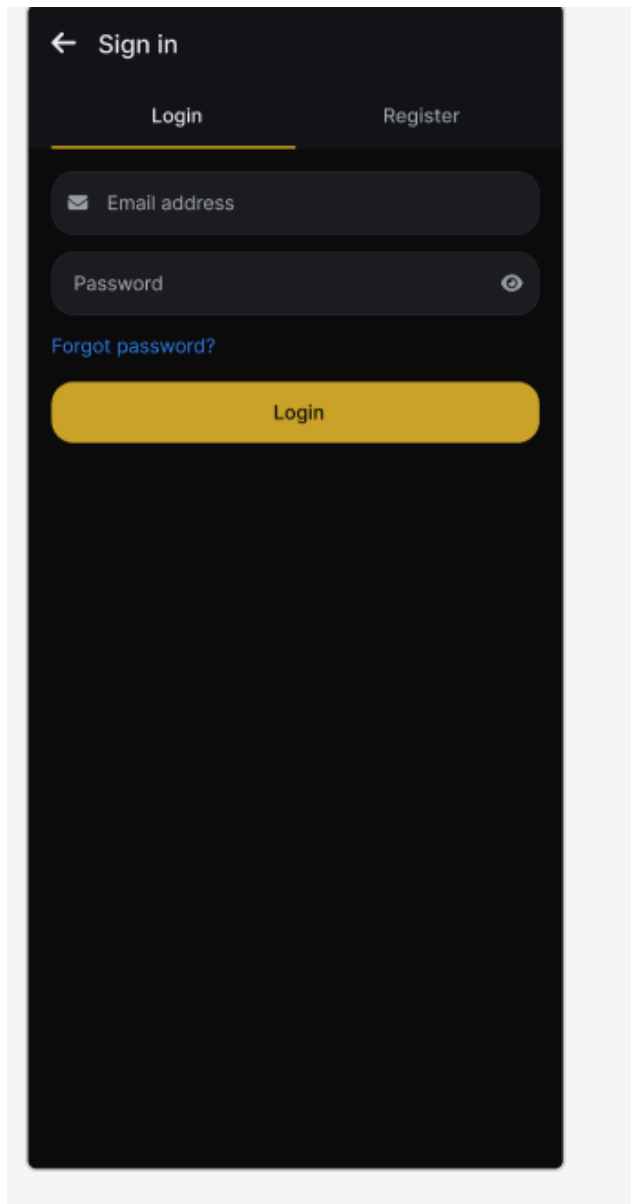
- Daily bias — Get updated market direction insights every day.
- Rule-gated alerts — Only trade when strict conditions align.
- 1-tap journal — Log and analyze trades instantly.

Authentication options are presented clearly:

- Continue with Google (white button with Google logo).
- Continue with Email (black button with envelope icon).
- Create account link for new users.

At the bottom, links to Privacy and Terms are shown for compliance. Error feedback (e.g., “Sign-in failed”) appears as a red banner for clarity.

Email Auth:



This screen allows users to sign in or register using their email credentials. The design follows the app’s dark theme with gold accents, maintaining brand consistency.

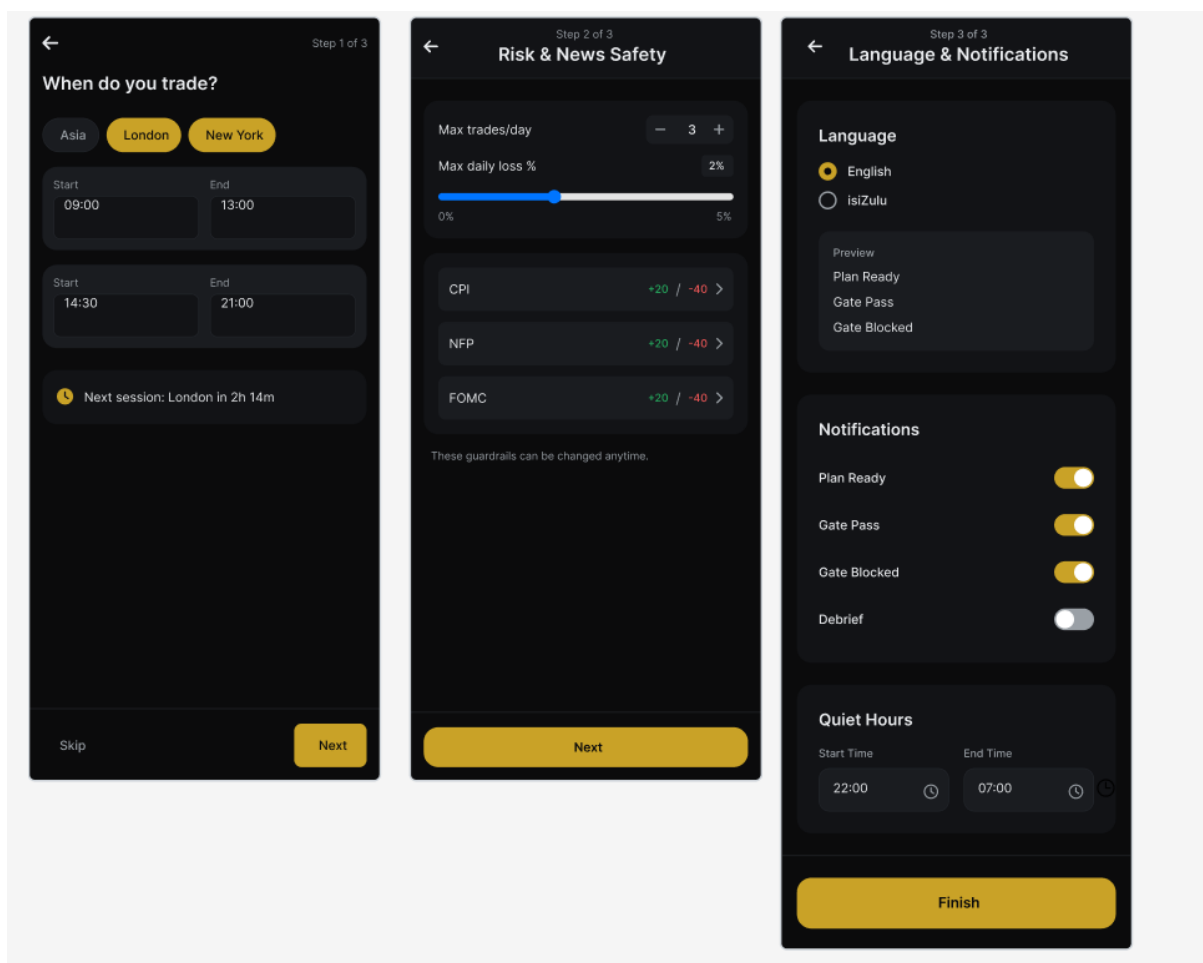
- At the top, a tab selector lets users switch between Login and Register.
- The Login form includes two input fields:
 - Email address (with envelope icon).
 - Password (with eye toggle for show/hide).
- A “Forgot password?” link is provided for recovery.

- The Login button is highlighted in gold (#FFD700), ensuring strong visibility and call-to-action.

For new users, the Register tab mirrors this layout but replaces the action button with Create account, also styled in gold.

This screen ensures secure authentication with email/password, and is integrated with Firebase/REST API for encryption and validation.

Onboarding A/B/C (sessions; risk & news mute; language & notifications):



Onboarding A – Trading Sessions

This screen lets the user define when they usually trade by selecting from Asia, London, or New York sessions.

- Users can specify start and end times for up to two sessions.
- A session countdown is displayed (e.g., *Next session: London in 2h 14m*).
- Navigation controls: Skip (bottom-left) and Next (bottom-right, gold button).

Onboarding B – Risk & News Safety

This screen helps configure risk guardrails and event filters.

- Users set maximum trades per day and maximum daily loss % using sliders and counters.
- High-impact news events (CPI, NFP, FOMC) are listed with potential impacts (+20 / -40).
- Guidance note: *“These guardrails can be changed anytime.”*
- Next button in gold continues the flow.

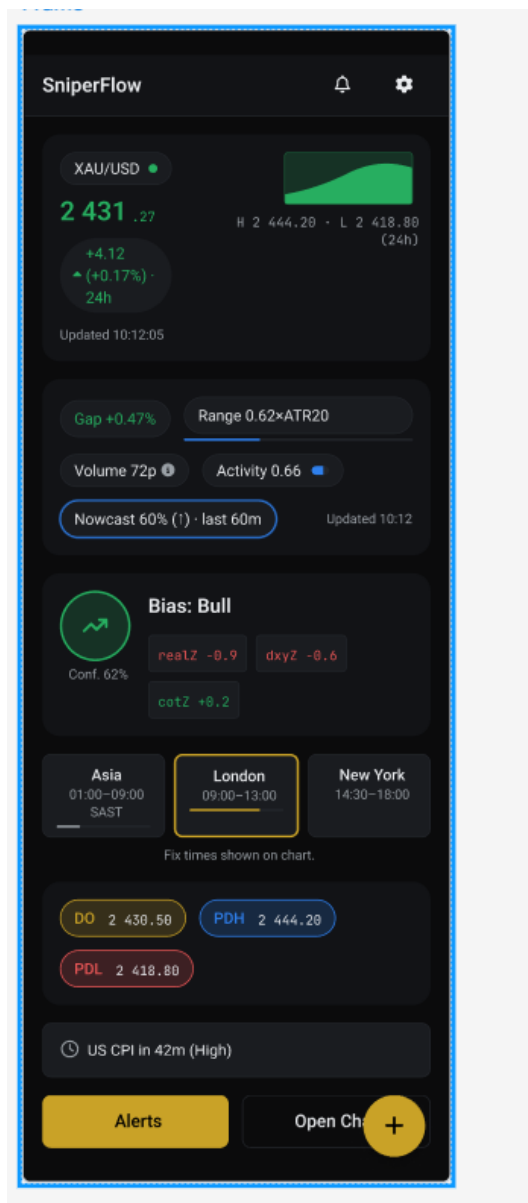
Onboarding C – Language & Notifications

This screen allows users to personalize language and notification preferences.

- Language: toggle between English and isiZulu.
- Notifications: switches for Plan Ready, Gate Pass, Gate Blocked, and Debrief.
- Quiet Hours: set start and end times to mute notifications.
- Finish button in gold completes onboarding.

These three onboarding screens together make the app adaptive, safe, and personalized ensuring users start with risk controls, language choice, and smart notifications already in place.

Home: live price, 24h delta, Market State, Nowcast ring, Sessions, Levels (DO/PDH/PDL), next high impact event.

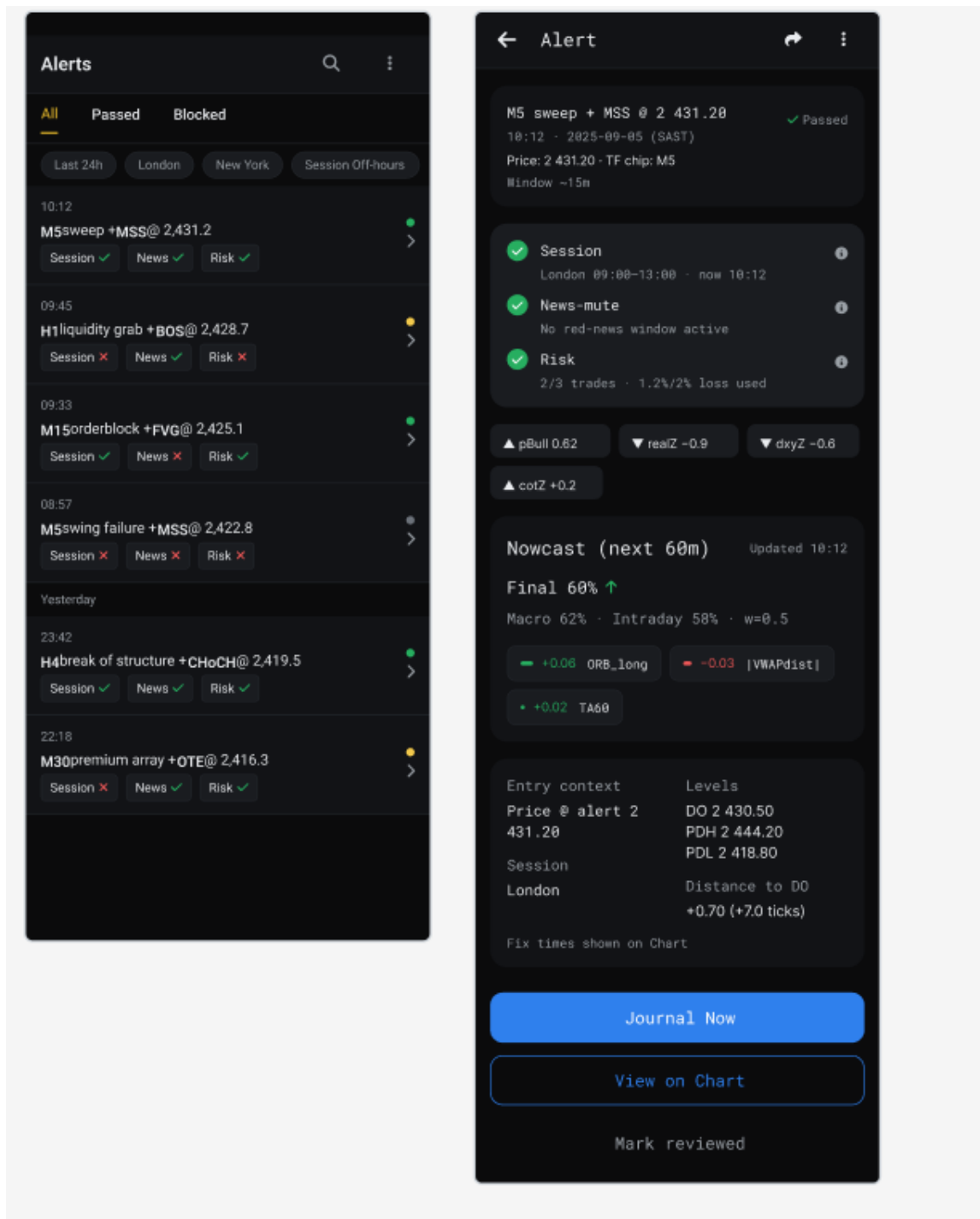


The SniperFlow Home screen serves as the trader's real-time dashboard for XAU/USD, combining live data, market bias, key levels, and upcoming events into a single actionable view.

- **Live Price Panel:**
At the top, the current XAU/USD price is shown with absolute and percentage change, plus a 24h high/low mini chart. Timestamp indicates last refresh.
- **Market Metrics:**
Displays Gap %, ATR20 range, Volume %, and Activity score.
A Nowcast indicator highlights intraday probability (e.g., *Nowcast 60% – last 60m*).
- **Bias Panel:**
Shows the current directional bias (Bull or Bear) with:
 - Confidence %

- Supporting drivers (e.g., *realZ*, *dxyZ*, *cotZ*) to make the bias explainable.
- Session Tracker:
Cards for Asia, London, and New York sessions with time windows in SAST.
Active session is highlighted.
- Key Levels:
Displays intraday reference points (DO, PDH, PDL) with exact price levels.
- Economic Event Countdown:
Shows the next major news release (e.g., *US CPI in 42m – High impact*).
- Quick Actions:
Bottom navigation buttons:
 - Alerts (yellow button for fast access)
 - Open Chart
 - Floating yellow “+” to add items such as journal entries or custom alerts.

Alerts screens:



Alerts List Screen

The Alerts list is designed to help traders quickly scan, filter, and prioritize signals.

- Tabs at the top: All, Passed, Blocked, with quick filters (Last 24h, London, New York, Session Off-hours).
- Each alert entry displays:

- Time of signal (e.g., 10:12, Yesterday 23:42).
 - Signal type and trigger (e.g., *MS sweep + MSS, H1 liquidity grab*).
 - Tag badges such as *Session, News, Risk* for context.
 - Status indicator: Passed  or Blocked .
- List layout ensures alerts can be reviewed in chronological order for both current and past days.

Alert Detail Screen

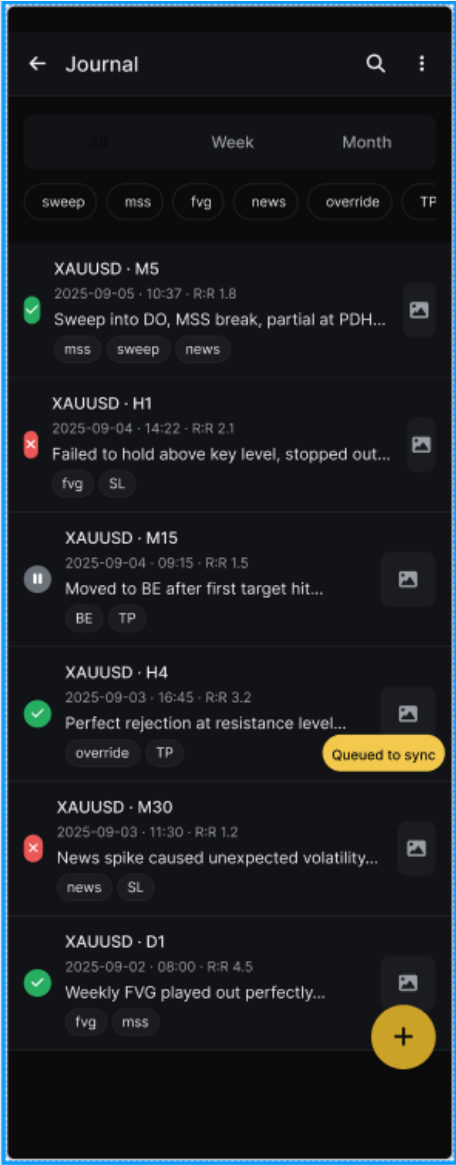
The detail screen provides full transparency into a single alert's conditions and reasoning.

- Header: Signal type, price, timestamp, timeframe (e.g., *M5 sweep @ 2431.20*).
- Session & Risk Checks:
 - Current session validity (London active, Asia closed).
 - News-mute status (whether red news blocked the trade).
 - Risk check (e.g., *2/3 trades used, 1.2%/2% daily loss*).
- Bias Context: Shows supporting bias drivers (e.g., *realZ, dxyZ, cotZ*) with values in green/red.
- Nowcast Panel: Intraday probability forecast (e.g., *Final 60% ↑*), split into Macro and Intraday components.
- Entry Context: Price, session, levels (DO, PDH, PDL) with distance to nearest key level.
- Actions:
 - Journal Now (directly log this trade setup into the journal).
 - View on Chart (open context visually).
 - Mark reviewed to confirm acknowledgment.

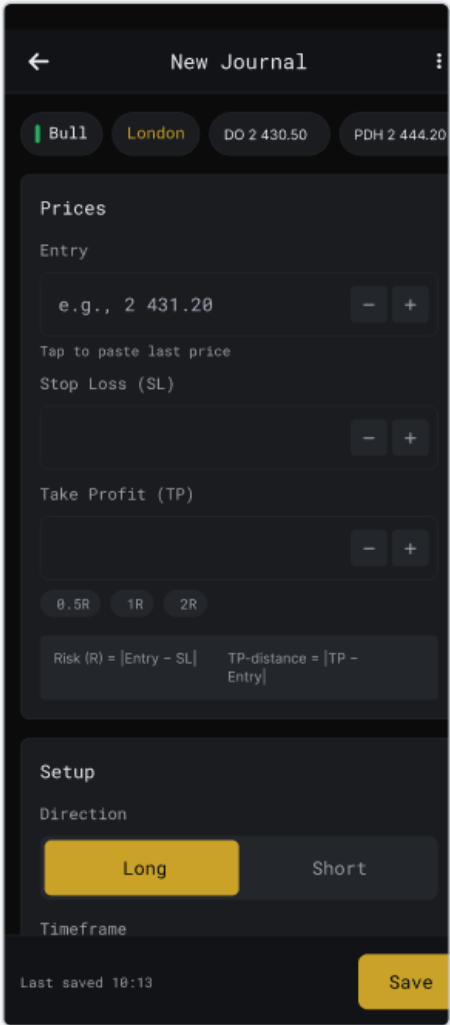
This design makes the Alerts system the heartbeat of SniperFlow signals are rule-gated, contextualized, and explainable, not just raw notifications. Traders can immediately act (journal or chart) or dismiss with full awareness of risk and bias conditions.

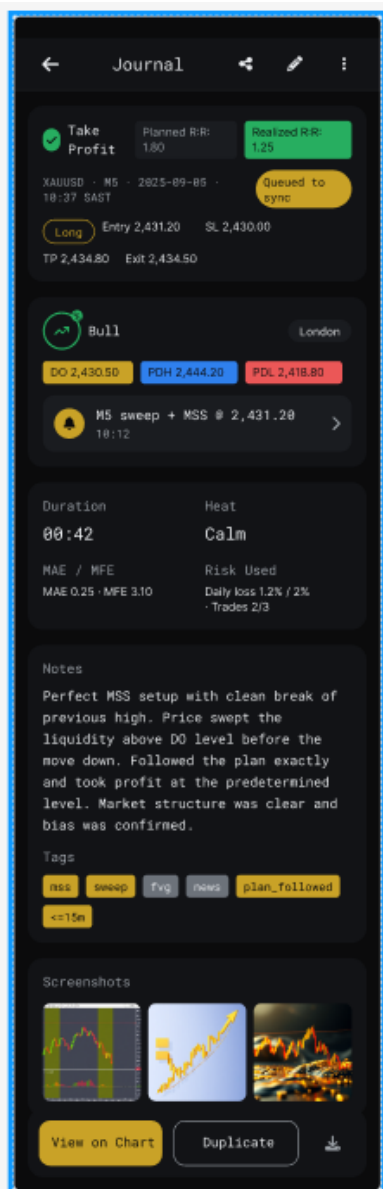
Chart:

Journal:



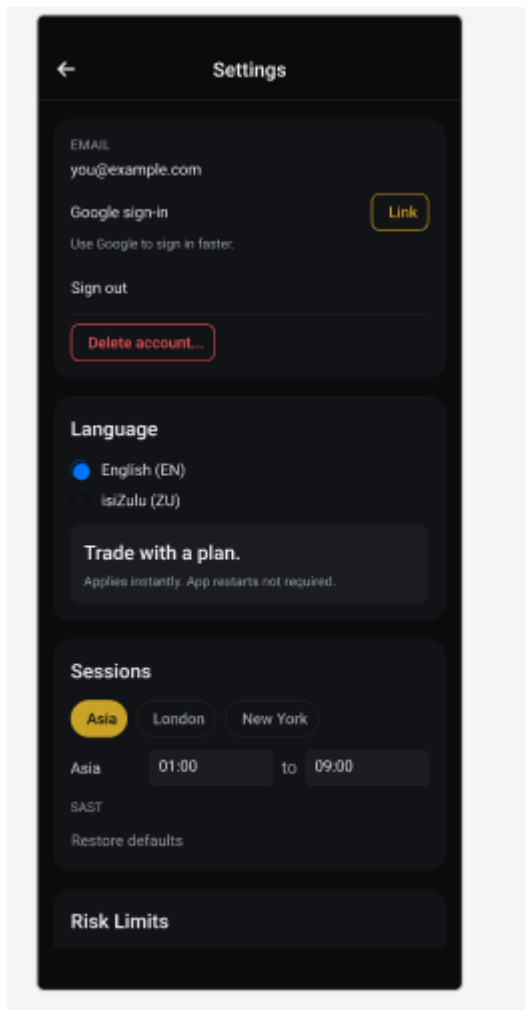
Frame





List (search, tags), New/Edit (validation for Long/Short, quick TP at 0.5R/1R/2R), Detail (R:R, notes, images, linked alert).

Settings:



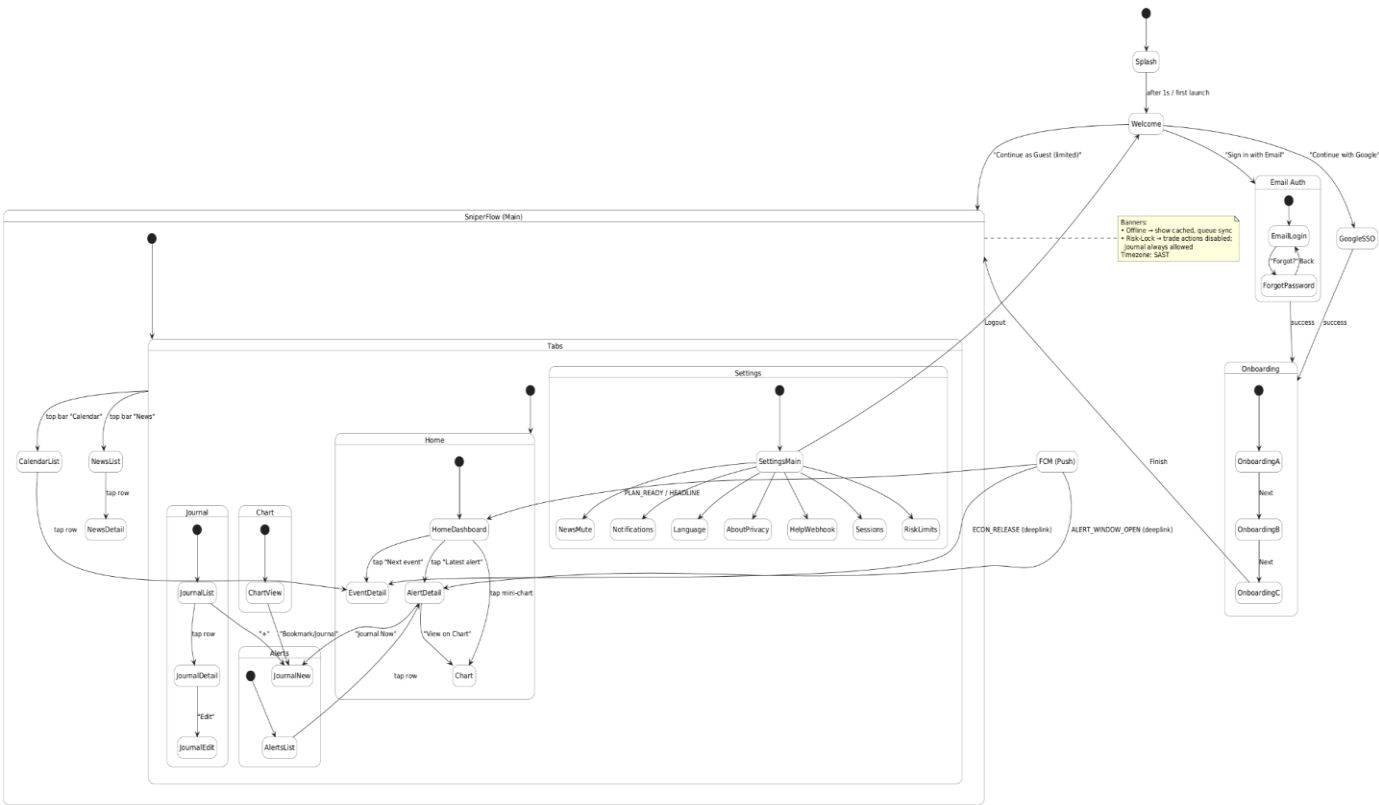
Account/SSO, Language, Sessions, Risk, News mute, Notifications, Data & About. • About/Privacy screens.

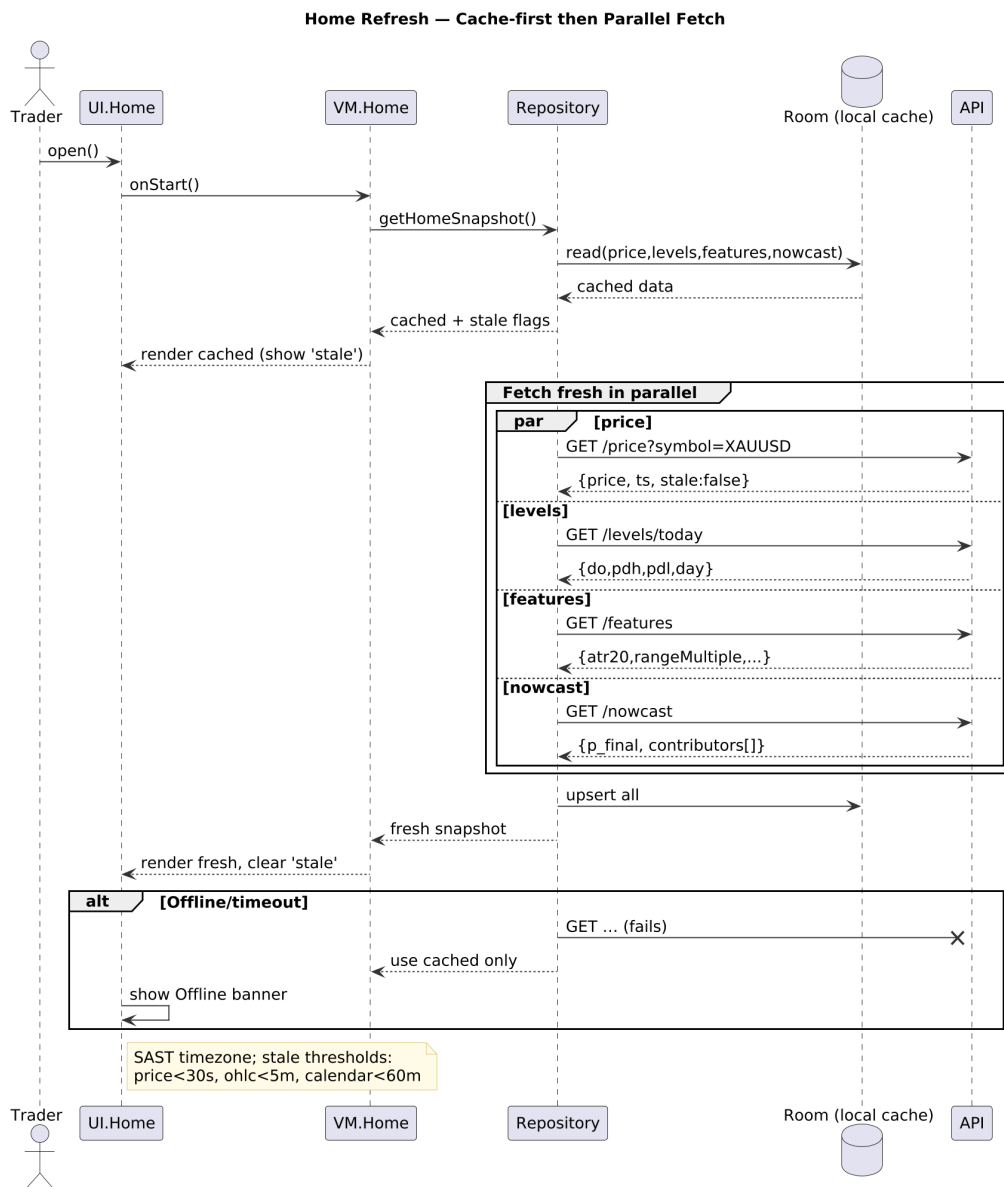
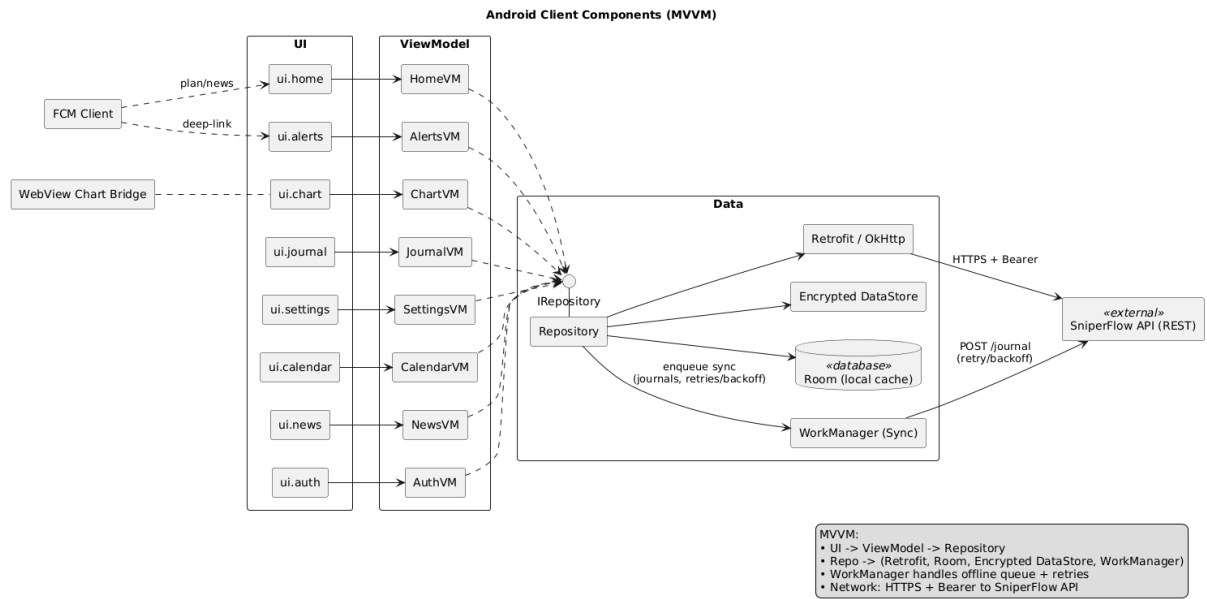
Calendar (Today/Week, impact/tag filters, countdowns) and Event Detail (Actual vs Forecast vs Previous, surprise/z, tilt) , News feed with sentiment and explain sheet:

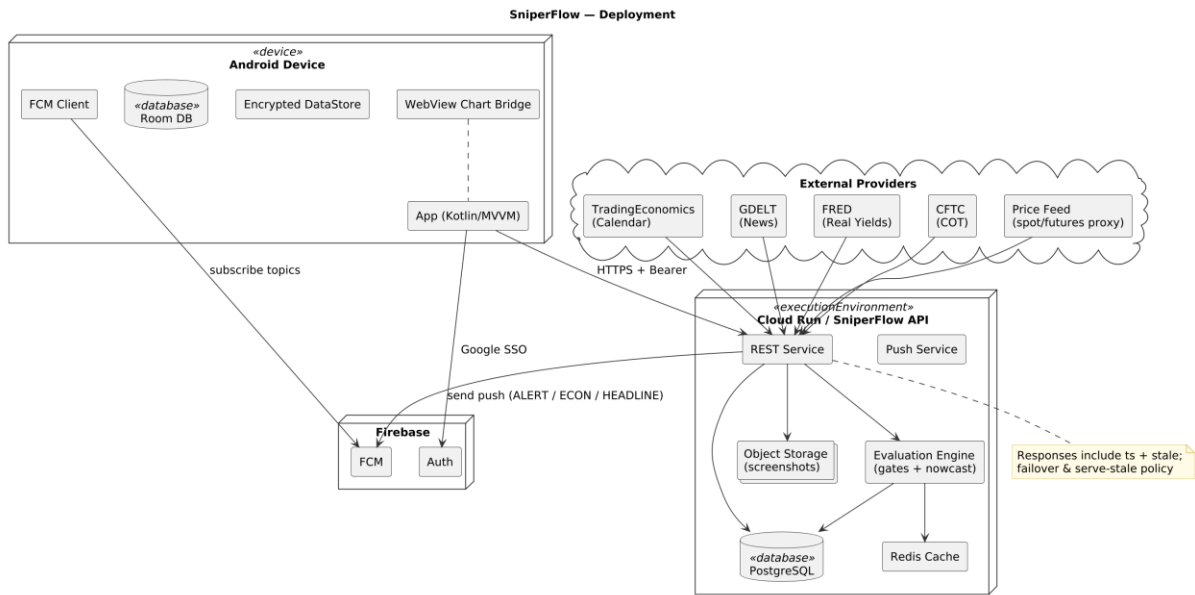
Navigation diagram (ASCII) Welcome → [Google | Email] → Onboarding A → B → C → Home Home ↔ Alerts ↔ Chart ↔ Journal ↔ Settings Home → Calendar → Event Detail Home → News → Headline Detail Alerts → Alert Detail → [Journal Now | View on Chart] Journal List → [New/Edit] → Journal Detail Settings → [About |

Privacy]

SniperFlow - Navigation (Screen Map)





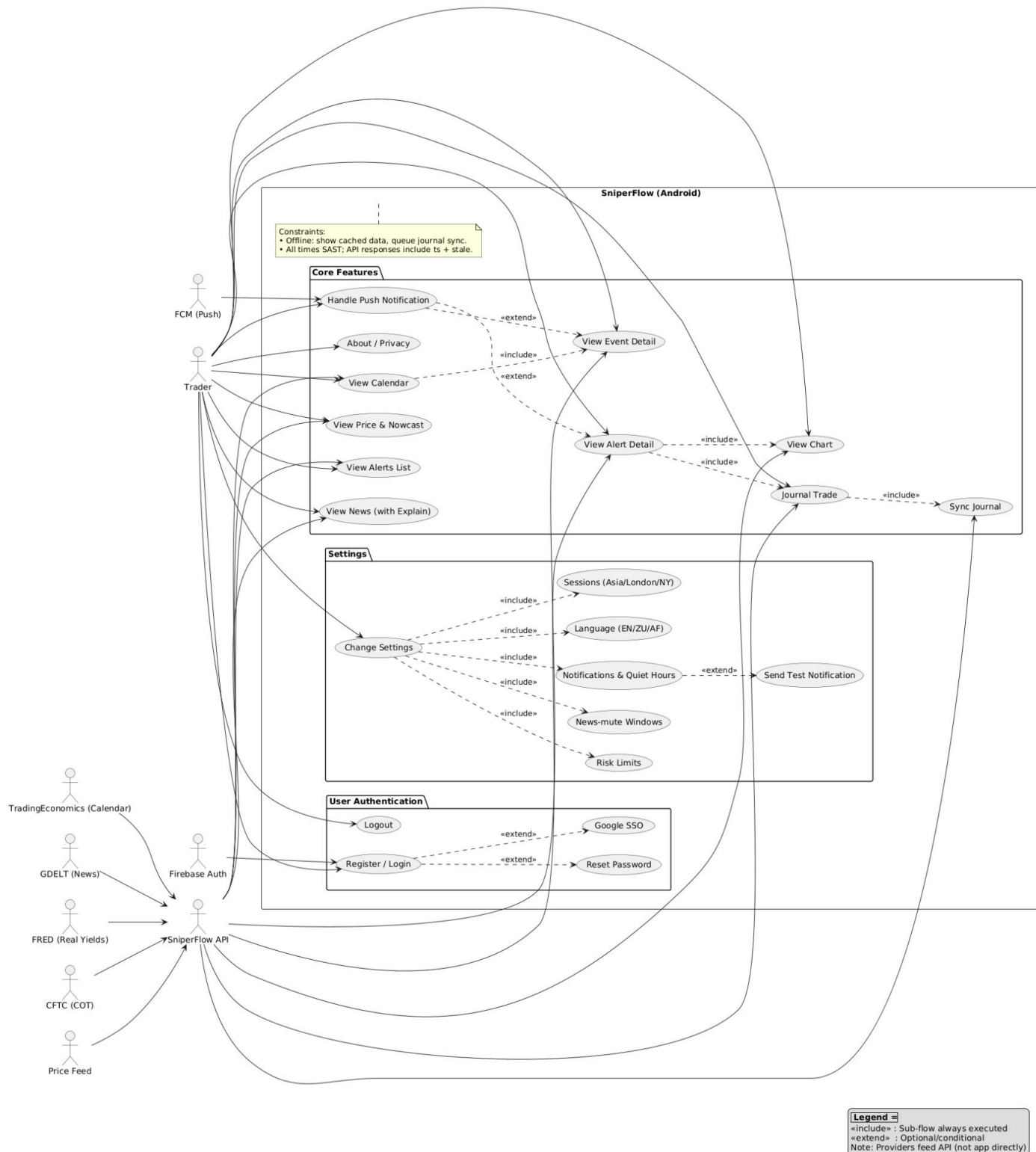


API payload examples (JSON) GET /nowcast { "p_now": 0.62, "p_macro": 0.55, "p_final": 0.58, "w": { "macro": 0.5, "intraday": 0.5 }, "contributors": [{ "name": "CPI surprise", "delta": -0.05 }, { "name": "London momentum", "delta": +0.03 }], "ts": 1693908720 } GET /calendar/release/{id} { "id": "us_cpi_yoy_2025_09", "timeSast": "2025-09-10T14:30:00+02:00", "impact": "High", "forecast": 3.2, "previous": 3.3, "actual": 3.4, "surprise": +0.2, "zScore": +0.7, "tiltDelta": -0.06 } Local storage schema (Room)

- Journal(id TEXT PK, ts LONG, tf TEXT, direction TEXT, entry REAL, sl REAL, tp REAL NULL, rrPlanned REAL NULL, rrRealized REAL NULL, outcome TEXT, note TEXT NULL, tags TEXT[], images TEXT[], alertId TEXT NULL, synced BOOL)
- Alert(id TEXT PK, ts LONG, tf TEXT, status TEXT, drivers JSON, gate JSON)
- Ohlc(tf TEXT, t LONG PK, o REAL, h REAL, l REAL, c REAL)
- Levels(day TEXT PK, do REAL, pdh REAL, pdl REAL)
- Prefs(key TEXT PK, value TEXT) Security
- Auth tokens stored via EncryptedSharedPreferences. TLS for all API calls. No secrets shipped in the app; API keys remain server side. Notifications & quiet hours
- Before posting a notification, the client checks quiet hours and suppresses non system channels.

UML

SniperFlow — Use Cases (Android client focus)



SniperFlow — Use Cases (Android client). User goals are modeled inside the system boundary. External actors (Firestore Auth, FCM, SniperFlow API, and data providers via the API) show integrations. «include» denotes mandatory sub-flows (e.g., Alert Detail includes Chart/Journal; Calendar includes Event Detail). «extend» denotes conditional flows (e.g., Login extends Google SSO; Push extends deep-links

to Alert/Event). Offline behavior (serve cached, queue journal sync) and SAST timebase are stated as constraints.

Normalizers (/calendar,/news,/nowcast) Auth (JWT) + DB Project plan (Gantt) See figure on next page. Milestones: research (D1–D2), contracts (D1–D4), scaffolding (D3–D10), Retrofit/data (D6–D12), offline/sync (D7–D14), calendar/news (D8–D15), notifications (D11–D16), localization (D12–D17), testing (D15–D22), store assets & privacy URL (D18–D23), polish (D24–D27). Buffer remains to D28. Conclusion This plan satisfies all POE minimums—email auth + SSO, user settings, REST API with database, Offline mode with sync, push notifications, and multi language support—and extends them with a gold specific Nowcast, gated alerts, explainability, and a disciplined journaling flow.

Figure 1 — 28 Day POE Gantt Schedule assumes start on 05 Sep 2025 (SAST). Critical path: Data layer → Offline/Sync → Calendar/News → Notifications → Testing.

